Describe a wave

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg sample with c=500 J/(kg·°C) warms 5°C. Find absorbed energy. If it is 5% of source energy, find source energy.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: f = cycles/Δt; T = 1/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

Worked reasoning: A wave makes 6 cycles in 3 s: f=2 Hz and period T=0.5 s. Amplitude measures maximum displacement, not the spacing of crests.

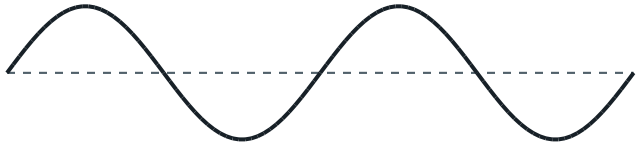
Your turn: Draw one spatial wave and label amplitude and wavelength 0. Separately mark one period on a displacement-time graph.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Label amplitude and one wavelength on the wave diagram. Explain why crest-to-trough height is not the amplitude.



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A point on a vibrating string completes 15 cycles in 0.60 s. Its crest-to-trough displacement is 0.08 m. Find frequency, period and amplitude. Sketch a displacement-time trace and label one period and one amplitude.



Marzano check - circle: ① ② ③

Describe a wave

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Continue down each column.

05 Explain your reasoning

Does increasing amplitude necessarily increase frequency? Use the definitions to explain 0.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 0.4 kg sample absorbs 2400 J while warming 6°C. Find specific heat and state a necessary assumption. Label temperatures, samples and relevant energy transfers.

Sample A

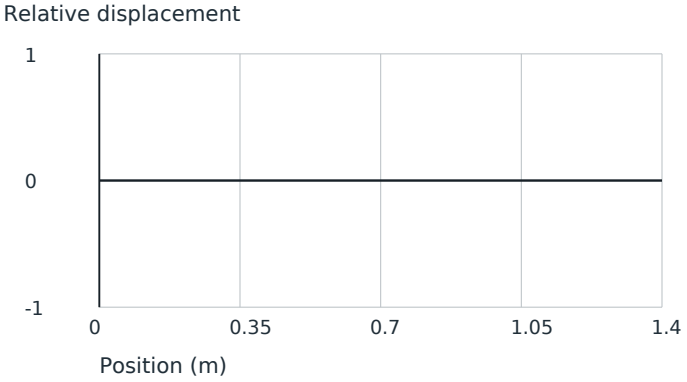
Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A sinusoidal snapshot has crests at x=0.20,0.70, 1.20 m. Find λ. If T=0.10 s, find f.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A point completes 15 cycles in 6 s. Successive crests are 1.2 m apart. Find period, frequency and wave speed.



Marzano check - circle: 1 2 3